

Lab 17: Capstone Systems Synthesis

BIOL-1

Name: _____ Date: _____

Learning Objectives

By the end of this lab, you should be able to:

1. Connect cell structure, genetics, evolution, and ecology in one causal explanation.
2. Trace how molecular changes can scale up to organism, population, and ecosystem effects.
3. Use evidence from multiple course units to support a biological claim.
4. Identify feedback loops and tradeoffs in a biological system.
5. Draw a capstone systems blueprint that connects at least four levels of biology with labeled flows of matter, energy, and information, including at least one feedback loop, and defend it against peer critique.

Overview

This capstone lab asks you to build a systems explanation instead of memorizing separate units. You will follow one scenario from molecule to ecosystem and show how the course ideas fit together.

Materials

- Pencil or pen
- Two sheets of blank paper (one for your blueprint draft, one for a partner's swap review)
- This worksheet

Scenario

A drought changes plant growth in a local ecosystem. Some plant individuals carry a genetic variant that changes water-use efficiency. Herbivores, predators, decomposers, and nutrient cycling all respond over time.

Part 1: Scale Map

Complete the table with one effect at each scale.

Scale	What changes in the scenario? Evidence or reasoning
Molecule / gene	
Cell	
Organism	
Population	
Community	
Ecosystem	

Part 2: Causal Chain

1. Write a six-step causal chain from genetic variation to ecosystem-level change.

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2. Where could natural selection act in this scenario? Identify the trait, environment, and fitness difference.

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3. Where could feedback occur? Describe one reinforcing or balancing feedback loop.

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Part 3: Evidence Check

Choose two course modules from Modules 01-16 and explain how each helps you reason about the drought scenario.

Module	Concept used	How it supports your explanation
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Part 4: Systems Blueprint

4. Plan before you draw. Before any drawing, list the **at least four levels** you will connect (choose from: molecule, organelle, cell, tissue, organism, population, community, ecosystem) and describe each arrow you plan to draw — what flows along it (matter, energy, or information) and in which direction.

5. Draw your blueprint. Now draw **one causal loop diagram** that matches your plan.

Requirements:

- At least **four levels** from the list above, each written as a labeled node (a simple circle or box).
- Every arrow labeled with what it carries: **matter**, **energy**, or **information** (e.g. "energy" from plants to herbivores; "information" from DNA to enzyme).
- At least **one feedback loop** — a closed cycle of arrows — marked with + (reinforcing: change grows) or – (balancing: change is damped).

6. Rank the tradeoffs. Every level of biology pays a cost. Rank each tradeoff **1–3** (1 = most important to the drought scenario, 3 = least), then justify your top-ranked tradeoff in one line.

Tradeoff

Rank (1–3) One-line justification

Fast growth vs. cancer risk (cells)

Reproduction now vs. survival later (organisms)

Individual fitness vs. population stability (populations / ecosystems)

7. Peer defense. Swap blueprints with a partner. Find the **weakest arrow** in their loop — the link you find least convincing — and either defend or attack it in **two sentences** (write about *their* diagram; then they do the same for yours).

Capstone Claim

8. Make one claim about how life works as a system. Support it with at least three ideas from different parts of BIOL-1.